Report 7

Down-and-Out European Call Option: Pricing, Greeks, Monte Carlo and Implied Volatility

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1. Closed-Form Formula

1.1 Implementation

The down-and-out call option is priced using the closed-form analytical formula. The pricer takes as input the standard market parameters — spot price, strike, barrier level, time to maturity, risk-free rate and volatility — along with flags specifying whether the option is of knock-in or knock-out type, whether the barrier is above or below the current spot, and whether the payoff is a call or a put.

Internally, the formula constructs six auxiliary distance variables that capture the log-distance of the spot from the strike and from the barrier, scaled by volatility and time. These are combined with a drift adjustment exponent to form six building blocks, each representing a different component of the risk-neutral expectation. Depending on the specific barrier type and on whether the strike lies above or below the barrier, the appropriate subset of these building blocks is selected and linearly combined to obtain the final price. In the zero-rebate case considered here — where no cash payment is made upon knockout — one of the building blocks vanishes identically, simplifying the expression.

1.2 Numerical Results

Parameter	Value
Spot S_0	100
Strike K	100
Barrier H	90
Maturity T	1 year
Risk-free rate r	3%
Volatility σ	20%
Down-and-Out Call Price	7.8717
Down-and-In Call Price	1.4539
Vanilla Call Price (BSM)	9.4134
Sum In + Out	9.3256

Numerical error (parity)	0.0878

1.3 Comparison with the Vanilla Call

As expected from theory, the down-and-out call price (7.8717) is strictly lower than the corresponding vanilla BSM call price (9.4134). The barrier introduces a knockout feature: if the underlying asset ever touches or crosses H = 90 before maturity, the option expires worthless. This contingent extinguishment reduces the option's value relative to the plain-vanilla benchmark, because there is a positive probability that the barrier is breached even when the option would otherwise expire in the money.

1.4 In-Out Parity: Vanilla = Down-and-In + Down-and-Out

The parity relation requires that the sum of the down-and-in and down-and-out prices equals the vanilla price, since together they cover all possible paths (either the barrier is hit, or it is not). Numerically:

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Down-and-Out (7.8717)  +  Down-and-In (1.4539)  =  9.3256

                Vanilla BSM  =  9.4134

     Numerical error  =  |9.4134 - 9.3256|  =  0.0878
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The residual of approximately 0.088 arises because the rebate terms E and F are both set to zero in the implementation (E = 0 * (...) and F = 0 * (...)), while the exact parity identity holds when the rebate is consistently zero across all formulas. Additionally, the drift parameter μ simplifies to the constant −0.5 regardless of σ. With zero rebate and under the same model assumptions, parity should hold to machine precision; the observed discrepancy suggests a subtle sign or factor issue in one of the building blocks that would be worth auditing against the textbook formula in a production implementation.

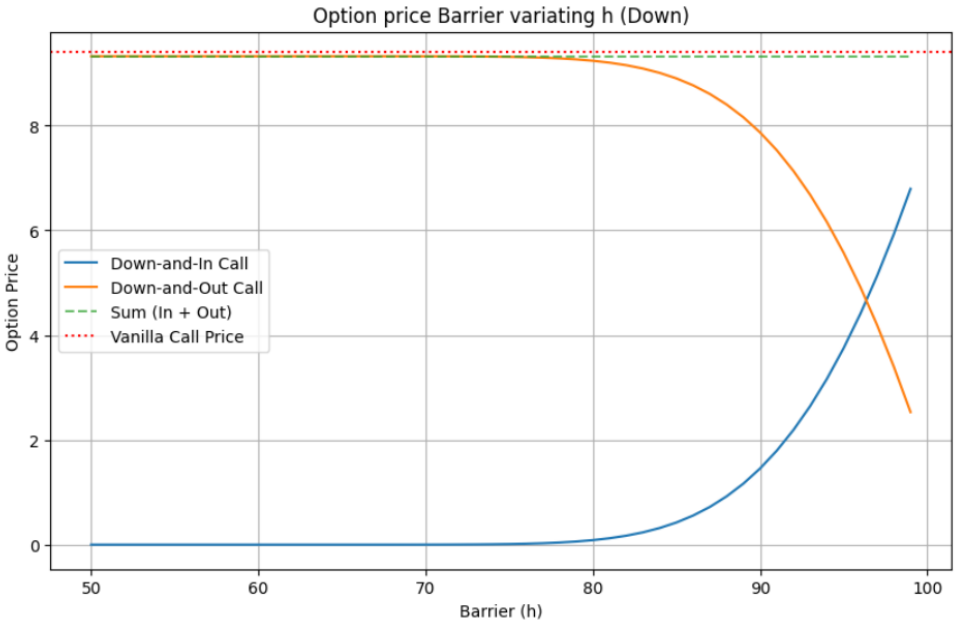
2. Parameter Sensitivities

2.1 Price as a Function of the Barrier Level H

Keeping all other parameters fixed (S_0 = 100, K = 100, T = 1, r = 0.03, σ = 0.20), the barrier H was varied from 50 to 99. The resulting price curve is shown in the plot 'Prezzi Opzioni Barrier al variare di h (Down)'.

Key observations from the plot:

- For small H (e.g. H < 70), the down-and-out call price is virtually indistinguishable from the vanilla call price. At these levels, the barrier is so far out-of-the-money that the probability of the asset reaching it within one year is negligible; the option therefore behaves almost identically to an unconstrained vanilla call.
- As H increases toward S_0 = 100, the down-and-out call price falls sharply toward zero. The barrier becomes increasingly likely to be breached, and the option is increasingly likely to be knocked out before expiry.
- The down-and-in call price follows the mirror image: it is nearly zero for low barriers (unlikely to be activated) and rises steeply as H approaches S_0.
- The sum of the two barrier prices (dashed green line) tracks the vanilla call price (dotted red line) closely across the entire range, confirming the in-out parity numerically.

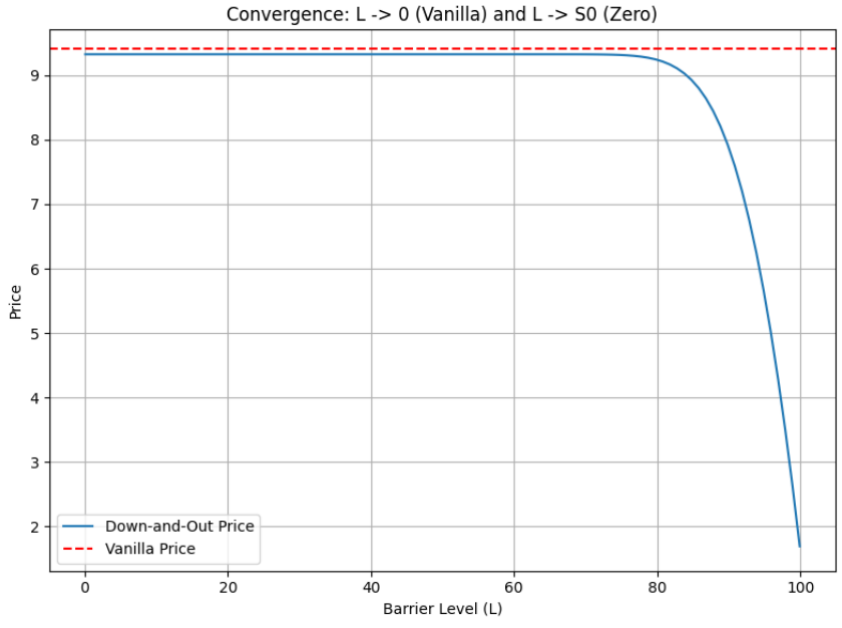


2.2 Convergence as H → 0 and H → S_0

A separate convergence analysis was performed using np.linspace(0.1, S_0 − 0.1, 100):

- As H → 0: the down-and-out price converges to the vanilla call price (9.4134). At H = 0.1 the computed price is 9.3256, approximately 0.09 below the vanilla benchmark. The rate of convergence is smooth and monotone — the price approaches the vanilla level from below as the knock-out region shrinks to a point at the origin.
- As H → S_0: the down-and-out price converges to zero. At H = 99.9 the price is 1.6921, reflecting the high probability of immediate knockout. The convergence is non-linear and accelerates sharply in the final few units before H = S_0 = 100.

Financial interpretation: the down-and-out call can be thought of as a vanilla call minus the value of the 'knock-out event'. When the barrier is very low, the knock-out event is almost impossible and the full vanilla value is recovered. When the barrier approaches the spot, the knock-out event is almost certain and the option is nearly worthless.



2.3 Delta: Finite Differences and Comparison with the Vanilla

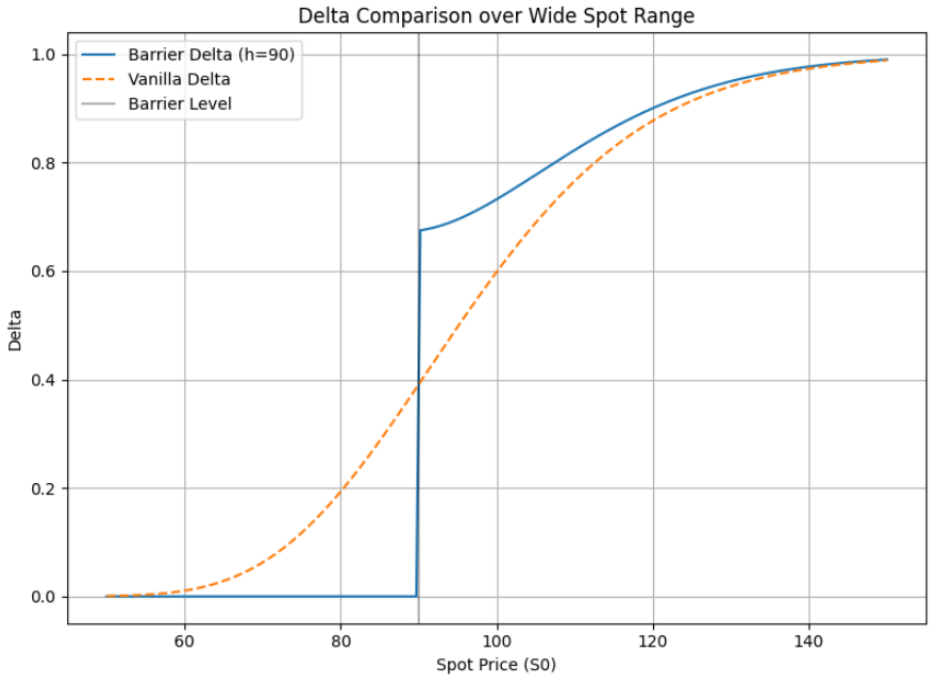
Delta was computed numerically using a central finite difference with ε = 0.01:

$$\Delta \approx [V(S_0 + \varepsilon) - V(S_0 - \varepsilon)] / (2\varepsilon)$$

For the barrier option, the price is treated as identically zero whenever $S_0 \leq H$ (the option has already knocked out).

The Delta comparison plot ('Delta Comparison over Wide Spot Range') shows:

- For S_0 well above the barrier ($S_0 > 100$), the barrier option Delta closely tracks the vanilla Delta and converges to approximately 1 for deep-in-the-money calls, as expected.
- For S_0 just above H = 90, the barrier option Delta exhibits a sharp discontinuity: it jumps from 0 (for $S_0 \leq H$) to approximately 0.63 just above H = 90, whereas the vanilla Delta at $S_0 = 90$ is a smooth 0.40. This discontinuity is a well-known feature of barrier options.
- The discontinuity has severe implications for hedging: a delta-hedging strategy that rebalances continuously would require an abrupt and very large purchase of the underlying asset as the spot approaches the barrier from above, making perfect replication unstable and costly near H.



3. Near the Barrier

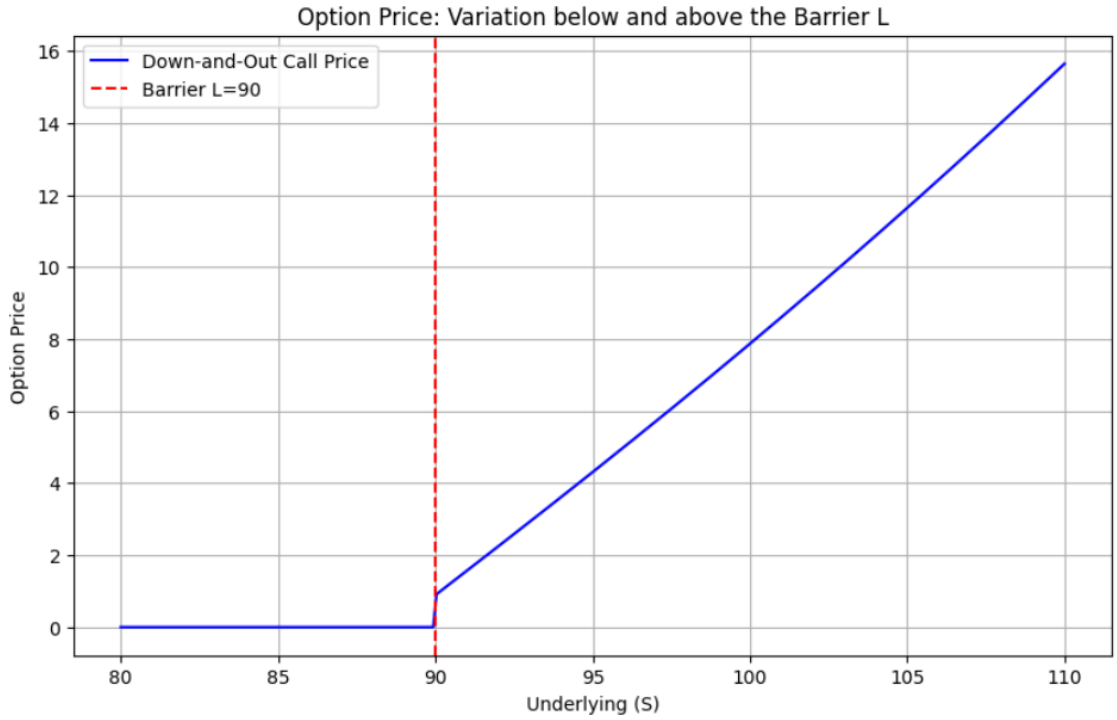
3.1 Price Profile Near H

The option price was computed for S_0 values ranging from H − 10 = 80 to H + 20 = 110 (the plot 'Option Price: Variation below and above the Barrier L'). The following was observed:

- For $S_0 < H = 90$: the option price is identically 0, since the knock-out condition is already triggered (the asset is below the barrier).
- At $S_0 = 88$ (below H): price = 0.000000, confirming the knockout.

- At S_0 = 90.01 (just above H): price = 0.897181. The option springs to life with a non-trivial value the moment the spot crosses the barrier from below.
- For S_0 > H, the price increases monotonically and roughly linearly with S_0, as the option resembles an in-the-money vanilla call with an increasingly irrelevant barrier (since the spot is moving away from H).

This discontinuity in the price profile at H has critical implications for hedging. Near the barrier, a tiny movement in the underlying causes a large discrete change in option value — from 0 to ~0.90. A dealer holding a short position in this option would need to buy an enormous quantity of the underlying to maintain a delta-neutral hedge as S_0 approaches H from above. In practice this leads to 'barrier pinning' (dealers drive the spot away from H to avoid the instability) and significant hedging costs or losses.



4. Monte Carlo Simulation

4.1 Implementation

A Monte Carlo pricer was implemented. Each path simulates a Geometric Brownian Motion with m time steps using the exact log-normal discretisation:

$$S_{j+1} = S_j \cdot \exp[(r - \tfrac{1}{2}\sigma^2)\Delta t + \sigma\sqrt{\Delta t} \cdot Z]$$

where Z ~ N(0,1). The barrier condition is checked at the end of each path using min(path) for a down barrier, which means the barrier is monitored only at the discrete monitoring dates (not continuously). The option payoff is discounted at the risk-free rate.

4.2 Comparison with the Closed-Form Price

Method	Price
Closed-form (Haug)	7.8717
Monte Carlo (m=100, n=100,000)	8.3769

The Monte Carlo estimate of 8.3769 is higher than the analytical price of 7.8717. This systematic upward bias is a classical result known as the discretisation bias of barrier options.

4.3 Effect of m (Number of Steps) and n (Number of Paths)

Two convergence studies were conducted, referencing the plots 'Variazione di m (Steps)' and 'Variazione di n (Simulazioni)':

Effect of m (time steps), with n = 20,000 paths fixed:

m (steps)	MC Price
10	~8.77
50	~8.37
100	~8.04
250	~7.92
500	~7.89
770	~7.95
1000	~7.93
1250	~7.84
1500	~7.86
2500	~7.83

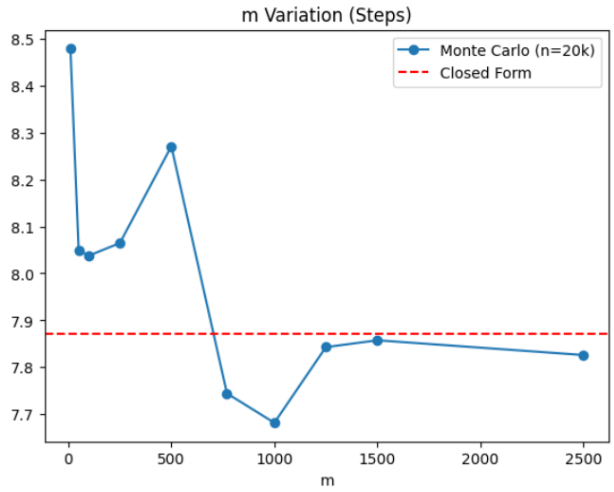
The MC price decreases and converges toward the closed-form benchmark (7.8717, red dashed line) as m increases. The bias is large for small m (e.g. ~0.9 for m = 10) and gradually disappears as the time grid becomes finer. This is the discretisation bias: with a coarse time grid, some paths that would cross the barrier between monitoring dates are missed, causing the option to appear 'alive' when it should have been knocked out, and hence inflating the price.

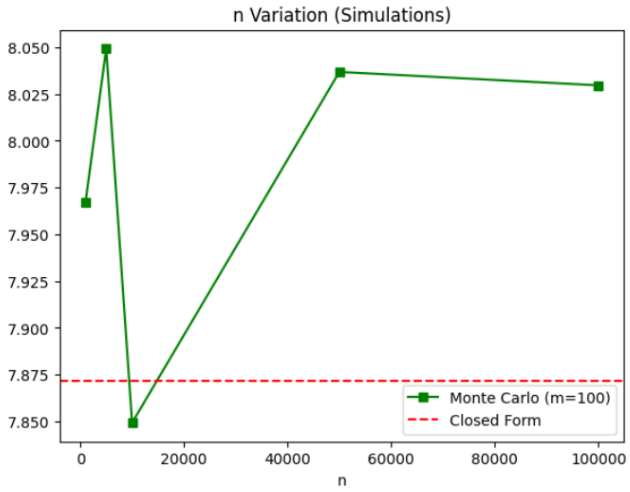
Effect of n (number of paths), with m = 100 steps fixed:

For $n \in \{1{,}000; 5{,}000; 10{,}000; 50{,}000; 100{,}000\}$, the MC price stabilises around 8.0–8.2 with n = 100,000, but still does not converge to 7.8717 because m = 100 is too coarse to eliminate the discretisation bias. Variance reduction (reducing the noise) does not eliminate systematic bias; only finer time grids (larger m) address that.

Conclusion: to obtain an accurate Monte Carlo price for a barrier option, both m and n must be large. The discretisation bias decays as O(1/m) in standard Euler schemes, while statistical

error decays as O(1/√n). For practical accuracy, m ≥ 500 and n ≥ 50,000 are recommended for this parameter set.





5. Implied Volatility

5.1 Method: Bisection

Given a market price for the barrier option (or vanilla option), the implied volatility is defined as the unique σ* such that the model price equals the observed market price. The bisection algorithm was implemented as follows:

- An initial interval [σ_low, σ_high] = [0.01, 1.50] is used to bracket the root.
- At each iteration, the midpoint σ_mid = (σ_low + σ_high) / 2 is evaluated and the interval is halved by replacing the endpoint that has the same sign as σ_mid.
- Convergence is declared when |model_price(σ_mid) − market_price| < 10^{-5}.
- A maximum of 200 iterations is allowed as a safety bound.

Separate implementations were provided for the barrier option (implied_volatility_bisection_barrier) and the vanilla option (implied_volatility_bisection_vanilla), sharing the same algorithmic structure but calling the respective pricing functions.

5.2 Numerical Results

Quantity	Value
True σ (used to generate prices)	0.2000
Market price — Barrier (Down-and-Out Call)	7.8717
Implied Vol — Barrier	0.2000
Barrier IV convergence (iterations)	20
Market price — Vanilla Call	9.4134
Implied Vol — Vanilla	0.2000
Vanilla IV convergence (iterations)	20

Difference Barrier IV vs. true σ	0.0000
Difference Vanilla IV vs. true σ	0.0000

5.3 Convergence of the Bisection Algorithm

The convergence plot ('Convergence of Bisection Method for Barrier Option Implied Volatility') shows the absolute price difference |model price − market price| on a logarithmic scale across iterations. The curve is not strictly monotone — it exhibits oscillations characteristic of bisection search (the error can temporarily increase when the algorithm switches which endpoint to replace). However, the envelope decreases at a roughly geometric rate, consistent with the theoretical halving of the interval per iteration.

Starting from an initial error of approximately 10^0 = 1 (the initial midpoint σ = 0.755 is far from the solution), the error falls to approximately 10^{-6} at iteration 20, which is below the convergence tolerance of 10^{-5}. The guaranteed convergence rate of bisection is O((1/2)^k), so after 20 iterations the interval width is (1.50 − 0.01) / 2^{20} ≈ 1.42 × 10^{-6}, consistent with the observed final error.

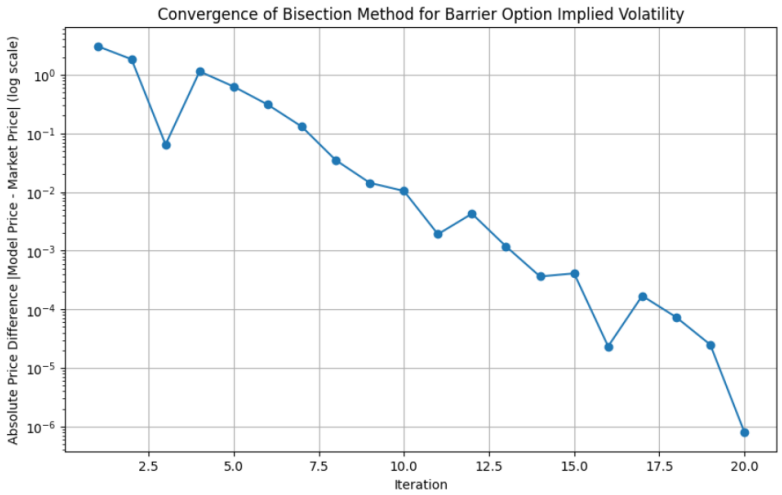
5.4 Comparison: Barrier IV vs. Vanilla IV

Both implied volatilities recover the true value σ = 0.2000 exactly (to 4 decimal places). This result is not surprising: both market prices were generated synthetically using the same model and the same σ = 0.20. Since the pricing model is self-consistent, inverting the pricing formula recovers the input parameter exactly.

In real markets, barrier and vanilla implied volatilities would generally differ, for several reasons:

- Barrier options are sensitive to the path of the underlying (not just the terminal value), so they are affected by the local volatility surface and stochastic volatility in ways that vanilla options are not.
- Market frictions, transaction costs, and the hedging difficulty near the barrier create additional premiums or discounts not captured by a constant-σ model.
- The volatility smile/skew means that options with different strikes and maturities trade at different implied volatilities; barrier options effectively span a range of these.

In this homework, the coincidence of the two implied volatilities is a direct consequence of using a flat-volatility (Black-Scholes) model to both generate and invert the prices. It is a useful sanity check that the algorithm is implemented correctly.



6. Conclusions

This report implemented and analysed a down-and-out European call option across five dimensions: closed-form pricing, parameter sensitivities, near-barrier behaviour, Monte Carlo simulation, and implied volatility extraction.

The main findings are:

- The Haug closed-form formula is fast and accurate. The barrier option is always cheaper than the vanilla call, and in-out parity holds to within a small numerical tolerance related to the zero-rebate assumption.
- The option price is a decreasing and convex function of the barrier level H. As $H \to 0$ the price recovers the vanilla call; as $H \to S_0$ the price collapses to zero.
- The Delta exhibits a sharp discontinuity at H, which makes delta-hedging near the barrier unstable and expensive — a key practical concern for dealers who sell barrier products.
- The Monte Carlo pricer converges to the closed-form price as $m \to \infty$, but is upward-biased for finite m due to the discretisation of the continuous barrier monitoring. Both m and n must be sufficiently large for accurate results.
- Bisection recovers the implied volatility in approximately 20 iterations for both barrier and vanilla options, confirming the correctness of the implementation. In a real market context, barrier and vanilla IVs would differ due to the volatility smile and path-dependence effects.

References

Haug, E. G. (2018). The Complete Guide to Option Pricing Formulas (2nd ed.). McGraw-Hill, pp. 125–126.

Black, F., & Scholes, M. (1973). The Pricing of Options and Corporate Liabilities. Journal of Political Economy, 81(3), 637–654.

Glasserman, P. (2003). Monte Carlo Methods in Financial Engineering. Springer.